

AL DHAFRA, United Arab Emirates

WMO: 412162

Lat: 24.248N

Lon: 54.548E

Elev: 76

StdP: 14.66

Time Zone: 4.00 (E04)

Period: 01-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a) 1	(b) 51.6	(c) 53.6	(d) 28.9	(e) 23.0	(f) 88.8	(g) 33.5	(h) 28.1	(i) 89.6	(j) 22.5	(k) 71.9	(l) 20.5	(m) 70.9	(n) 4.2	(o) 230	(p) 0.453

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a) 8	(b) 23.6	(c) 114.9	(d) 73.6	(e) 112.9	(f) 73.5	(g) 111.0	(h) 74.3	(i) 86.7	(j) 96.1	(k) 85.5	(l) 95.3	(m) 84.4	(n) 94.8	(o) 8.6	(p) 310

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
84.4	181.3	92.7	82.8	171.7	92.2	82.1	167.4	92.0	51.7	97.6	50.3	94.5	48.8	94.5	90.9

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a) 21.9	(b) 19.4	(c) 17.3	(e) N/A	(f) N/A	(g) N/A	(h) N/A	(i) N/A	(j) N/A	(k) N/A	(l) N/A	(m) N/A	(n) N/A	(o) N/A	(p) N/A		
			DB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A		
			WB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	83.6	66.4	69.7	74.7	83.6	90.3	93.9	97.6	99.3	93.8	86.4	77.6	69.4
	DBStd	12.06	3.90	5.47	6.31	5.26	4.76	3.97	4.21	3.07	3.33	3.31	4.07	3.68
	HDD50	0	0	0	0	0	0	0	0	0	0	0	0	0
	HDD65	48	28	11	2	0	0	0	0	0	0	0	0	7
	CDD50	12280	508	552	766	1007	1250	1317	1477	1529	1315	1130	828	601
	CDD65	6853	71	143	303	557	785	867	1012	1064	865	665	378	143
	CDH74	99408	565	1246	3055	7047	11819	13819	16874	17698	13470	9049	3795	971
	CDH80	65680	109	475	1442	4102	7870	9644	12417	13234	9211	5339	1635	202
Wind	WSAvg	7.7	6.9	8.5	8.7	8.4	8.3	8.5	8.5	8.2	7.1	6.2	6.2	6.5
Precipitation	PrecAvg	2.3	0.4	0.4	0.5	0.2	0.0	0.0	0.1	0.0	0.0	0.0	0.1	0.4
	PrecMax	5.7	2.1	3.5	2.3	2.1	1.0	0.6	0.8	0.9	0.1	0.2	0.8	2.6
	PrecMin	0.1	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0
	PrecStd	1.7	0.6	0.7	0.7	0.5	0.2	0.1	0.2	0.2	0.0	0.0	0.2	0.6
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	86.2	94.9	101.8	107.5	114.4	116.4	116.8	116.9	112.9	105.5	97.3	87.8
		MCWB	63.1	61.5	64.5	67.4	69.6	71.3	74.5	74.7	71.2	69.9	67.5	65.1
	2%	DB	82.7	90.1	96.5	104.1	111.1	113.1	114.7	114.8	109.6	102.3	94.6	84.5
		MCWB	62.3	63.2	64.9	66.7	70.4	72.1	75.0	74.0	72.3	70.3	67.3	64.5
	5%	DB	80.3	85.9	92.8	100.9	107.9	110.9	111.6	112.8	107.4	100.4	91.4	82.4
		MCWB	61.5	62.4	64.0	66.9	70.3	72.5	76.1	74.6	73.5	71.3	68.7	65.7
	10%	DB	76.9	82.1	88.1	98.2	105.4	107.6	109.2	109.7	104.4	98.2	89.2	79.2
		MCWB	61.2	61.6	63.8	66.7	70.8	73.5	76.6	75.8	75.2	71.8	68.5	65.1
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	68.3	70.0	71.8	74.5	81.9	84.6	88.2	88.1	87.1	82.6	77.5	71.3
		MCDB	75.5	78.6	82.3	87.1	97.4	95.2	98.7	97.6	94.0	91.4	84.6	79.1
	2%	WB	66.3	67.8	70.0	72.8	79.9	83.3	86.7	86.8	85.2	81.0	75.4	69.4
		MCDB	75.3	78.0	80.9	86.5	91.8	94.3	96.6	96.2	94.1	91.1	84.8	77.9
	5%	WB	64.8	66.1	68.8	71.5	78.3	82.1	85.5	85.6	83.9	79.9	73.7	68.1
		MCDB	74.5	76.3	80.3	85.7	91.3	93.5	95.6	95.5	93.1	90.2	84.1	77.1
	10%	WB	63.4	64.6	67.3	70.2	76.5	80.7	84.3	84.4	82.7	78.6	71.9	66.8
		MCDB	73.0	75.8	79.7	85.6	91.7	93.3	95.2	95.1	92.2	88.7	82.6	76.3
Mean Daily Temperature Range	5% DB	MDBR	20.9	21.7	23.1	25.4	28.0	27.8	23.4	23.6	24.1	24.6	21.5	20.6
		MCDBR	25.3	27.1	29.2	30.7	32.2	31.9	25.6	26.5	26.3	26.4	24.2	23.8
	5% WB	MCWBR	11.2	11.2	10.7	10.3	11.0	11.7	10.8	11.6	12.8	11.7	10.2	10.7
		MCWBR	21.3	21.6	21.7	25.3	28.3	26.9	22.1	22.6	24.1	24.2	21.5	20.9
Clear-Sky Solar Irradiance	taub	taub	0.450	0.503	0.570	0.609	0.596	0.637	0.731	0.705	0.600	0.505	0.459	0.441
		taud	1.999	1.856	1.704	1.653	1.668	1.623	1.513	1.558	1.710	1.891	2.012	2.031
	Ebn at Noon	Ebn at Noon	251	246	236	231	233	222	202	207	226	242	246	247
		Edh at Noon	50	62	76	81	80	83	92	88	74	59	49	47
All-Sky Solar Radiation	RadAvg	RadAvg	1352	1593	1866	2121	2344	2383	2225	2160	2019	1776	1449	1272
	RadStd	RadStd	71	115	120	124	63	51	74	48	40	31	34	72

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)
Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
Regional (7 neighbors)	+0.99	+1.31	N/A	N/A	N/A	N/A	N/A	N/A	+338	+335

Nomenclature: See separate page